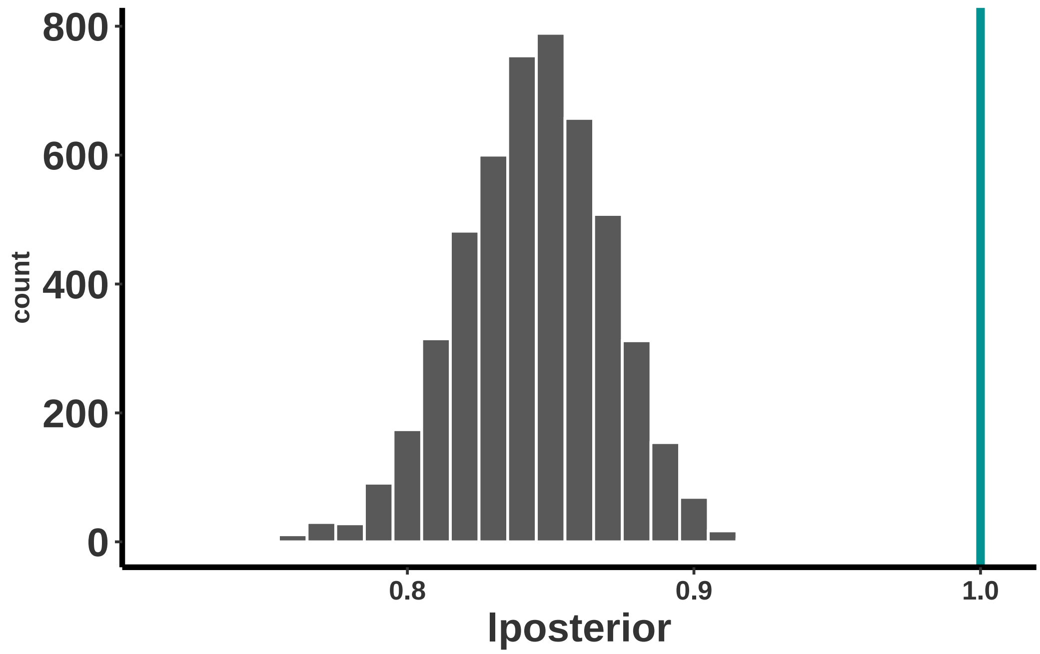
medianL	meanL	lcl	ucl	pincrease
0.845	0.844	0.791	0.892	0



Para concluir, la población de *Dactylorhiza incarnata* muestreado por Tamm desde 1944 a 1971 tiene un crecimiento poblacional menor que 1.0 (lambda promedio de 0.84), con un intervalo de confianza de 0.79 a 0.89. Esto indica que la población esta reduciendo en el tiempo por aproximadamente 16%, pero que realmente la reducción pudiese ser entre 21% a 11%. El índice de *pincrease* es el valor de *p* para evaluar si la población esta creciendo o decreciendo. En este caso el valor es <0.001, lo que indica que la población está decreciendo.

```
Dacty_0.1bdf = as.data.frame(Dacty_0.1b)
Dacty_0.1_summary <- dplyr::summarize(
  Dacty_0.1bdf,
  medianL = median(lposterior),
  meanL = mean(lposterior),
  lcl = quantile(lposterior, probs = 0.025),
  ucl = quantile(lposterior, probs = 0.975),
  pincrease = sum(lposterior > 1.) / n()
)
```

```
gt(signif(Dacty_0.1_summary, digits = 3))
```

Part II

Apéndices

Apéndice A

Lista completa de PPM en orquídea

En esta sección se hará recopilación de los estudios de dinámica poblacional en orquídeas que han usado matrices de transición de estados y de edades. Se identifica algunos de los temas principales de los trabajos.

En la siguiente tabla se presentan los estudios de dinámica poblacional en orquídeas que usaron PPM y la(s) preguntas principales que los autores intentaron responder. En la gran mayoría de los trabajos se evalúa múltiples hipótesis/ideas sobre la dinámica poblacional de las orquídeas. Para información especifica sobre la metodología usada en cada estudio, se recomienda revisar los trabajos originales.

- Clave de las variables
  - Riesgo: Si se evaluó el riesgo de extinción de la especie.
  - Variación espacial: Si se evaluó la variación espacial de la especie.
  - Variación temporal: Si se evaluó la variación temporal de la especie.
  - Latencia: Si se evaluó la importancia de la latencia (Dormancy) de la especie.
  - Métodos de estimación de parametros: Si se evaluó métodos alternos de estimación de parametros (no tradicional).
  - Historia de Vida: Si se evaluó la historia de vida de la especie en un contexto evolutivo.
  - Otro: Si se evaluó otra pregunta no incluida en las anteriores.
  - Referencia: Referencia del estudio.

```
library(readr)
library(tidyverse)
library(dplyr)
library(gt)
library(flextable)
library(ftExtra)
```

Genero	Especies	Riesgo	Var esp.	Var temp	Lat
Aspasia	*principissa*	Si	-	Si	-
Brasavolla	*cucullata*	Si	Si	Si	-
Broughtonia	*cubensis*	Si	Si	Si	-
Caladenia	*amoena*	Si	-	-	Si
Caladenia	*argocalla*	Si	-	-	Si
Caladenia	*clavigera*	Si	-	-	Si
Caladenia	*elegans*	Si	-	-	Si
Caladenia	*graniticola*	Si	-	-	Si
Caladenia	*macroclavia*	Si	-	-	Si
Caladenia	*oenochila*	Si	-	-	Si
Caladenia	*rosella*	Si	-	-	Si
Caladenia	*valida*	Si	-	-	Si
Caladenia	*orientalis*	Si	Si	Si	Si
Cephalanthera	*longifolia*	-	-	-	Si
Cleistes	*divaricata* var. *divaricata*	Si	Si	Si	Si
Cleistes	*divaricata* var. *bifaria*	Si	Si	Si	Si
Cleistes	*bifaria*	-	Si	-	Si
Corallorhiza	*trifida*	Si	-	Si	Si
Crepidium	*acuminatum*	Si	-	Si	-
Cypripedium	*acaule*	-	-	-	-
Cypripedium	*calceolus*	Si	Si	Si	Si
Cypripedium	*calceolus*	Si	Si	Si	Si
Cypripedium	*calceolus*	-	-	-	-
Cypripedium	*fasciculatum*	Si	Si	Si	Si
Cypripedium	*parviflorum*	-	-	-	Si
Cypripedium	*reginae*	-	Si	-	Si
Cypripedium	*lentiginosum*	-	-	-	-
Dactylorhiza	*hatagirea*	Si	Si	Si	Si
Dactylorhiza	*lapponica*	-	Si	Si	Si

Genero	Especies	Riesgo	Var esp.	Var temp	Lat
Dendrophylax	*lindenii*	Si	-	Si	-
Epidendrum	*xanthinum*	Si	Si	-	-
Epipactis	*atrorubens*	Si	Si	Si	Si
Epipactis	atrorubens	-	-	-	-
Erycina	*crista-galli*	Si	Si	-	-
Erycina	*crista-galli*	-	-	-	-
Guarianthe	*aurantiaca*	Si	Si	-	-
Herminium	*monorchis*	Si	-	Si	Si
Himantoglossum	*hircinum*	Si	-	Si	-
Himantoglossum	*hircinum*	Si	-	Si	-
Himantoglossum	*hircinum*	Si	-	Si	Si
Isotria	*medeoloides*	Si	-	Si	-
Jacquiniella	*leucomelana*	Si	-	Si	-
Jacquiniella	*teretifolia*	Si	-	Si	-
Laelia	*autumnalis*	Si	Si	Si	-
Laelia	*speciosa*	Si	-	Si	-
Lepanthes	*caritensis*	Si	Si	Si	-
Lepanthes	*caritensis*	Si	Si	Si	-
Lepanthes	*eltoroensis*	Si	Si	-	-
Lepanthes	*eltoroensis*	Si	Si	Si	-
Lepanthes	*eltoroensis*	Si	Si	Si	Si
Lepanthes	*rubripetala*	-	-	-	-
Lepanthes	*rubripetala*	Si	Si	Si	-
Lepanthes	*rupestris*	-	-	-	-
Lepanthes	acuminata	Si	-	-	-
Lycaste	*aromatica*	Si	-	Si	-
Neotinea	*ustulata*	Si	Si	-	Si
Oeceoclades	*maculata*	Si	Si	Si	-
Oncidium	*poikilostalix*	Si	-	-	-

Genero	Especies	Riesgo	Var esp.	Var temp	Lat
Orchis	*purpurea*	Si	-	Si	-
Orchis	*ustulata*	Si	Si	-	-
Phaius	*australis*	Si	Si	-	-
Platanthera	*hookeri*	Si	Si	-	-
Platanthera	*praeclara*	Si	Si	Si	Si
Platanthera	*macrophylla*	Si	Si	Si	-
Platanthera	*orbiculata*	Si	Si	Si	-
Prasophyllum	*correctum*	Si		Si	Si
Pseudorchis	albida	-	-	-	-
Rodriguezia	*granadensis*	Si	Si	Si	-
Serapias	*cordigera*	Si	Si	Si	Si
Spathoglottis	*plicata*	Si	Si	Si	-
Spiranthes	*delitescens*	Si	Si	Si	Si
Spiranthes	*parskii*	Si		-	-
Telipogon	*helleri*	Si	-	-	-
Tolumnia	*variegata*	Si	Si	Si	-
Trichocentrum	*undulatum*	-	-	-	-

Figura 25.6: *Trichocentrum undulatum.* Foto: Hong Liu

Notas para añadir a la tabla

- IPM - Integrated Population Model
- Transient dynamics
- Transfer function

Especies de orquídeas estudio sin PPM

En la siguiente lista incluimos algunos de los trabajos de dinámica poblacional pero que no usan PPM para responder a su pregunta. La lista es muy grande y no es exhaustiva. La lista incluye trabajos que usan otros métodos de estimación de parámetros, como modelos de regresión, o que usan IPM (Integral population models) pero no usan matrices de transición de estados o edades que es método alterno de estimación de parámetros que se usa en este libro.

Figura 25.7: *Spiranthes delitescens.* Foto: Mitchel McClaran

Genero	Especies	Otro	Ref
Isotria	*medeoloides*	no matrix	@alahuhta2017instant
Cyclopogon	*luteoalbus*	no matrix	@juarez2014viability
Cypripedium	*calceolus*	no matrix	@davison2013contributions
Cypripedium	*candidum*	no matrix	@shefferson2017predicting

Genero	Especies	Otro	Ref
Gymnadenia	conopsea	no matrix	@oien2002flowering
Ophrys	*sphegodes*	no matrix	@hutchings2010population
Ophrys	*sphegodes*	no matrix	@shefferson2017predicting
Orchis	*morio*	no matrix	@wells1998flowering
Spiranthes	*spiralis*	no matrix	@jacquemyn2007long

Apéndice B

Hojas de datos en el campo

Por: Raymond L. Tremblay

Librerías de R requeridas para el siguiente módulo

```
library(tidyverse)
library(gt)
```

Presentamos un ejemplo de la hoja de recolección de datos y algunas variables incluido y que debería estar preparada antes de ir al campo y estar imprimido sobre papel que es impermeable y que se puede escribir con lápiz. Esos ultimos puntos son bien importante ya que la gran mayoría de los bolígrafos son soluble en agua y si el papel se moja se puede perder sus datos. Considerando que estas hojas de datos pudiese ser usado como fuente de averiguar los datos por décadas en el futuro. **RECUERDA que no se puede regresar con una maquina de tiempo para recoger los datos**. Si usted tiene su hoja de datos en el laboratorio y se le cae la taza café y fue escrito con bolígrafo se puede perder los datos. No se arriesga!! **Usa lápiz**.

Antes de ir al campo es importante tener la lista completa de todos los individuos en la hoja de datos, ya que esto le ayuda asegurarse durante el muestreo que no se le olvida ningún individuo. *Datos olvidado no se puede recuperar*.

Puede usar papel Rite-in-Rain o papel de impresora normal. Si usa papel normal, se puede poner en una bolsa de plástico con un pedazo de cartón para que no se doble. Si se moja, se puede secar y no se pierde los datos. Mi sugerencia use papel **Rite in the Rain** y lápiz.

Ejemplo de Hojas de campo

Hoja en blanco para llevarse

anio	Número_de_Ind	Etapa	Cantidad_flores_abierta	Cantidad_capullo	Cantidad_Frutos	Nur
2023	23001	...	...	...	...	
2023	23002	...	...	...	...	
2023	23003	...	...	...	...	
2024	23001	...	...	...	...	
2024	23002	...	...	...	...	
2024	24003	...	...	...	...	
2024	24004	...	...	...	...	
2024	24005	...	...	...	...	

Número_de_Ind	Etapa	Cantidad_flores_abierta	Cantidad_capullo	Cantidad_Frutos	Numero_F
23001	p	NA	NA	NA	
23002	j	NA	NA	NA	
23003	A1	5	10	1	
23004	Ao	0	0	0	
23005	m	NA	NA	NA	
24006	p	NA	NA	NA	
24007	j	NA	NA	NA	
24008	p	NA	NA	NA	

```
Hoja_de_campo %>% gt()
```

Ejemplos de tablas de Datos

Usando ejemplo de Tremblay y su codificación

- p == **plantula**.
- j == **juvenil**
- A0 == **adulto non-reproductivo**
- A1 == **adulto reproductivo**
- M = **muerto**

Hoja llena al final del muestreo

```
Hoja_llena %>% gt()
```

Nota que solamente los individuos 23003 y 23004 son adultos, uno tiene flores, capullos y frutos y el otro nada. A todos los adultos hay que llenar la información, no dejarlo en blanco. Para las otras etapas, no es necesario llenar la mayoría de las columnas ya que para las plántulas y muertos no PUEDEN tener esas variables suplementaria, pero los juveniles, se puede llenar la cantidad de hojas y sus tamaño. Nota que el individuo 23005 falleció.

Almacenamiento de datos de forma digital

Esa misma estructura de recoger los dados puede ser utilizado para ponerlo en una hoja de MSExcel, Google Sheet o MacOS Numbers.

- Es muy importante que haya dos hojas, una para los datos y otra para la metadata.
- La metadata es la descripción de los datos, como se recogieron, donde, cuando, por quien, las unidades, objetivo de la investigación, etc.
- En la hoja de datos, cada columna tiene información de solamente UNA variable y cada fila tiene información de solamente UN individuo. No debería tener las unidades indicado en las celdas de la hoja de datos, sino en la metadata o en nombre de la columna.
    - La primera fila tiene los nombres de las variables y la primera columna tiene los nombres de los individuos.
    - Los nombres de las columnas no pueden tener espacios, ni caracteres especiales, ni acentos, ni tildes, como en **ñ**.
    - Los nombres de los individuos no pueden tener espacios, ni caracteres especiales, ni acentos, ni tildes, como en **ñ**, **é**, y **%**. Preferiblemente que la codificación sea un patrón alfanumérico.
- La hojas de datos preferiblemente se guarda en hoja de datos separados, como .csv, .xls, .xlsx, .ods, etc. La ventaja de usar .csv es que es un archivo de texto y se puede abrir en cualquier programa de hoja de datos. Ese formato .csv, *comma separated variable*, es uno de los más viejo para almacenar datos, y es compatible con casi todos los programa de estadística. **NUNCA** guarda sus datos en un formato de programas especializados como SPSS, STATA, JMP, SAS, porque típicamente no se puede abrir en otros programas, y si no tiene (o pierda) la licencias se puede que nunca los podrán abrir en el futuro (experiencia personal).

Revisión:

RLT: Enero 19, 2025

RLT: Junio 18, 2025

Apéndice C

En este apéndice se añade enlaces para los datos usado en el libro y algunos pdf.

Tendrán que bajar estos datos a su ordenador en la carpeta de trabajo (asumo RStudio) para que los ejemplos funcionen.

Capítulo de Crecimiento Poblacional

Los datos de crecimiento poblacional se pueden descargar desde el siguiente enlace:

Laelia speciosa población 1

1. Laelia_speciosa_1.csv
2. Laelia_speciosa_2.csv

Capítulo de Olaf Tamm

Los datos de Olaf Tamm se pueden descargar desde el siguiente enlace: Estos fueron digitalizado por RLT

1. Olaf_Tamm_1972

Algunos artículos de Tamm (no fácil de encontrar en la web)

- Tamm1948.pdf
- Tamm1956.pdf
- Tamm1972.pdf

Archivos para calcular LTRE/ERTV

La hoja de Excel para calcular ERTV sin tener que usar R:

- LTRE_ERTV.xlsx

Chapter 26

Agradecimiento

26.0.0.1 Librerías de R requeridas para el siguiente módulo

```
library(tidyverse)
#library(gt)
library(flextable)
library(ftExtra)
```

Chapter 27

Agradecimientos

Este libro es un resultado de trabajo en común y el esfuerzo de muchas personas. Aprecio a todas las personas que han contribuido a este libro, ya sea con fotos, revisiones, o ayudando a entrar datos en COMPADRE y especialmente a los colaboradores. La información resumida en este libro no habría sido posibles sin las cientos y miles de horas que los estudiantes y colegas han dedicado a sus investigaciones y publicaciones. Con los resultados presentado en los artículos científicos y tesis hemos podido comenzar a entender y apreciar la diversidad de patrones de la historia de vida de las especies de orquídeas y comenzar a evaluar como el estudio de la historia de vida puede ayudarnos a evaluar la el riesgo de la supervivencia de la poblaciones y especies por causa natural o efecto antropogénico. El futuro también se ve muy interesante en que podemos comenzar a evaluar patrones que pudiese describir como la historia de vida de la orquídeas se parece y difiere de otros grupos taxonómicos y a lo mejor describir un otro angulo de por que la familia de las Orquídeas es tan abundante.

27.1 Fotos

Aprecio las siguientes personas por haber compartido fotos

```
Fotos=tribble(~"Origen de las Fotos", ~Especie, ~ Dirección,
              "Diana Molina Ozuma", "*Erycina crista-galli*", "Departamento de Ecología
              "James D. Ackerman", "*Dactylorhiza maculata*, *D. sambucina*", "Departmen
              "Eduardo A. Pérez García", "*Laelia speciosa*", "Universidad Nacional Autó
              "Hans Jacquemyn", "*Orchis purpurea*", "Leuven Plant Institute, KU Leuven,
              "Giuseppe Pellegrino", "*Serapias cordigera*", "Ecology and Earth Sciences
              "Mitch McClaran", "*Spiranthes delistecens*", "Professor of Range Manageme
"Hong Liu", "*Trichocentrum undulatum*", "Department of Earth and Environment, Florida I
"Edwin Guevara", "*Lepanthes caritensis* \n *Lepanthes eltoroensis* \n *Lepanthes woodbu
"Mark Blackman", "*Vanilla planifolia*", "Avenida del IMAN111-601, Insurgentes-Cuiculco,
```

```
"Leonel López-Toledo", "*Laelia speciosa*", "Instituto de Investigaciones sobre los Recu
"Alonso Argüero", "*Psychilis karwinskii*", "Departamento de Ecología y Recursos Natural
"Claudia C. Guitiérrez", "*Cypripedium irapeanum*", "27 A Oriente 1812-1, Fraccionamient
              )
```

```
Fotos_out=Fotos %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")

Fotos_out
```

Origen de las Fotos	Especie	Direcció
Diana Molina Ozuma	*Erycina crista-galli*	Departa
James D. Ackerman	*Dactylorhiza maculata*, *D. sambucina*	Departm
Eduardo A. Pérez García	*Laelia speciosa*	Universi
Hans Jacquemyn	*Orchis purpurea*	Leuven
Giuseppe Pellegrino	*Serapias cordigera*	Ecology
Mitch McClaran	*Spiranthes delistecens*	Professo
Hong Liu	*Trichocentrum undulatum*	Departm
Edwin Guevara	*Lepanthes caritensis* *Lepanthes eltoroensis* *Lepanthes woodburyana*	Marylan
Mark Blackman	*Vanilla planifolia*	Avenida
Leonel López-Toledo	*Laelia speciosa*	Instituto
Alonso Argüero	*Psychilis karwinskii*	Departa
Claudia C. Guitiérrez	*Cypripedium irapeanum*	27 A Ori

27.2 Revisores

Las siguientes personas ayudaron a revisar algunos de los capítulos. Todos los errores son míos y no de ellos.

```
Revisores=tribble(~"Nombre", ~"Institución",
              "Joel Tupac Otero", "Departamento de Ciencias Biológicas, Universidad Naci
              "Maria Eglée Pérez", "Departamento de Matemática, Universidad de Puerto Ri
              "Hibraim Adán Pérez Mendoza", "Facultad de Estudios Superiores-Iztacala UN
              )
```

```
Revisores_out=Revisores %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")

Revisores_out
```

Nombre	Institución
Joel Tupac Otero	Departamento de Ciencias Biológicas, Universidad Nacional de Colombia, Sede
Maria Eglée Pérez	Departamento de Matemática, Universidad de Puerto Rico, San Juan, Puerto Ri
Hibraim Adán Pérez Mendoza	Facultad de Estudios Superiores-Iztacala UNAM, México

---

27.3 Entrada de datos en COMPADRE

Aprecio las siguientes personas por haber entrado datos de PPM de orquídeas en COMPADRE

```
Benevolos=tribble(~"Nombre", ~"Institución",
              "Javielise Donato", "Universidad de Puerto Rico-Humacao, Puerto Rico",
              "Diana Clausia Molina Ozuna", "Conservación de la Biodiversidad, El Colegi
              "Nayliz M. Ortiz Ortiz", "Universidad de Puerto Rico-Humacao, Puerto Rico"
              "Gabriela Torres Torres", "Universidad del Valle, Colombia")
```

```
Benevolos_out=Benevolos %>%
  flextable() %>%
  theme_zebra() %>%
  theme_box() |>

  set_table_properties(layout = "autofit")
```

```
Benevolos_out
```

Nombre	Institución
Javielise Donato	Universidad de Puerto Rico-Humacao, Puerto Rico
Diana Clausia Molina Ozuna	Conservación de la Biodiversidad, El Colegio de la Frontera Sur, ECOSUR, Méxic
Nayliz M. Ortiz Ortiz	Universidad de Puerto Rico-Humacao, Puerto Rico
Gabriela Torres Torres	Universidad del Valle, Colombia

Acácio V, Holmgren M, Jansen PA, Schrotter O. 2007. Multiple recruitment limitation causes arrested succession in Mediterranean cork oak systems. *Ecosystems* 10:1220-1230.

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Ackerman J, Tremblay R, Pérez M, Madden H, Bechtold M, Boeken M. 2020. Small populations on small islands: What chance does an orchid have? *International Journal of Plant Sciences* 181:667-685.

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